



Apex Basics Assignment

Note: Both Basic and Intermediate Questions are compulsory to attempt.

Note: While submitting Assignment on Git Branch Name should be feature/apex-basic-assignment.

Write a Apex class with name **ApexBasicsAssignment** which will have following methods:

Basic Coding Questions

1. Create a function to check if a given string is a palindrome.
public static Boolean isPalindrome(String str)
2. Implement a function to find the sum of digits of a given integer.
Hint: Can use logic of converting Integer to string, then calculate the sum of each digit by looping on it.
public static Integer sumOfDigits(Integer num)
3. Write a function to reverse the elements of a list.
Note: Predefined functions are not allowed for this question
public static List<Integer> reverseList(List<Integer> inputList)
4. Create a function that will put all the values in the 2-d list into a map where the key should be the index of the row and the values which are stored with respect to the key.
Also, duplicate values inside the inner list should get removed as you can see in the below example.

Example:

Input: 2-d list: ((11,22), (111, 222, 222, 200))

Output: Map of Integer and 2-D list

Key	Value
0	(11,22)
1	(111, 222, 200)



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```
public static Map<Integer, List<Integer>> map2DList(List<List<Integer>>
inputList)
```

Intermediate Coding Questions

1. In the following list<list<String>> Remove the duplicate values using best practice. The Resulting string should be in sorted form.

For ex:

```
list<list<'a;b;a;v;c','x;x;y;c;q'>, list<'m;b;h;j;m','u;k;j;u'>>
```

Then the output should be:

```
list<list<'a;b;c;v','c;q;x;y'>, list<'b;h;j;m','j;k;u'>>
```

```
public static List<List<String>>
removeDuplicatesAndSort(List<List<String>> inputList)
```

2. Rotate the elements in the list:

a) Clockwise (towards the right) by 5

```
public static List<Integer> rotateClockwise(List<Integer> inputList, Integer
positions)
```

b) Anti-Clockwise (towards the left) by 7

```
public static List<Integer> rotateAntiClockwise(List<Integer> inputList,
Integer positions)
```

Note: If you can solve it in $O(n)$ complexity then you will get 5 marks extra for that.

3. Print all the LEADERS in the list. An element is a leader if it is greater than all the elements to its right side.

Sample list: <149, 1, 3, 248, 2, 8, 70, 5>

149 is not greater than 248 so it is not a leader, neither 1 nor 3 but 248 is, as it is greater than all the elements on its right, so is 70 & 5.



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So leaders are 248,70,5

```
public static List<Integer> findLeaders(List<Integer> inputList)
```